

Presuppositions (in DRT)

Week 10

Slides and materials based on the courses by
Noortje Venhuizen and Mareike Hartmann

Back to entailment

- Recall: sentence A entails sentence B ($A \models B$) iff whenever A is true, then B must also be true
- Entailment is a relation between the *propositions* expressed by the sentences A and B:
 - “John and Mary failed the test” $\models$ “Mary failed the test”
 - “John or Mary failed the test” $\models$ “someone failed the test”
 - “John is an intelligent student” $\models$ “John is a student”
 - “every student works” $\models$ “every blond student works”

More examples of entailment?

- *“the mathematician who proved Goldbach’s conjecture wasn’t a man”*
⊨? *“someone proved Goldbach’s conjecture”*
- *“Mary doesn’t love her husband”*
⊨? *“Mary has a husband” / “Mary is married”*
- *“it wasn’t Mary who broke the typewriter”*
⊨? *“somebody broke the typewriter”*
- *“John kissed every girl at the party”*
⊨? *“there were girls at the party”*

Entailment vs. Presupposition

- Entailment:

- “*Mary failed the test*” $\models$ “*Mary took the test*”
- “*Mary didn't fail the test*” $\not\models$ “*Mary took the test*”

- Presupposition:

- “*the mathematician who proved Goldbach’s conjecture was a woman*”
 >> “*someone proved Goldbach’s conjecture*”
- “*the mathematician who proved Goldbach’s conjecture wasn't a woman*”
 >> “*someone proved Goldbach’s conjecture*”

What are presuppositions?

- **Definition 1:** A presupposition of a statement is a proposition that must be true in order for the statement to be interpretable (to make sense) in the first place
- **Definition 2:** A presupposition is an implicit assumption about the world whose truth is taken for granted by the speaker

Back to definite descriptions

- Definite descriptions convey *uniqueness*:

“the chancellor decides”

$\mapsto$ *“there is exactly one chancellor, and they decide”*

$\mapsto \exists x(\text{chancellor}(x) \wedge \forall y(\text{chancellor}(y) \leftrightarrow x = y) \wedge \text{decide}(x))$

- *“the chancellor”* $\mapsto \lambda G. \exists x(\text{chancellor}(x) \wedge \forall y(\text{chancellor}(y) \leftrightarrow x = y) \wedge G(x))$
- *“the”* $\mapsto \lambda F \lambda G. \exists x(F(x) \wedge \forall y(F(y) \leftrightarrow x = y) \wedge G(x))$

Definite descriptions and uniqueness

- This is a problem for compositionality, e.g.:

“it is not the case that the chancellor decides”

- Compositional analysis of the sentence leads to:

$$\neg \exists x(\text{chancellor}(x) \wedge \forall y(\text{chancellor}(y) \leftrightarrow x = y) \wedge \text{decide}(x))$$

- *“Either there is no chancellor, or more than one, or there is exactly one chancellor and they do not decide”*

- But the correct representation should be:

$$\exists x(\text{chancellor}(x) \wedge \forall y(\text{chancellor}(y) \leftrightarrow x = y) \wedge \neg \text{decide}(x))$$

- *“There is exactly one chancellor, and they do not decide”*

Presupposition vs. Assertion

- A sentence (e.g. one containing a definite description) contains meaning information of (at least) two different types:
 - **Presupposition**: the requirements that the context must satisfy for the sentence to be interpretable
 - **Assertion**: the claims that are made (based on the context)

“the chancellor decides”

⊢ *“there is exactly one chancellor, and they decide”*

⊢ $\exists x(\text{chancellor}(x) \wedge \forall y(\text{chancellor}(y) \leftrightarrow x = y) \wedge \text{decide}(x))$

Presupposition projection

- Presuppositions are not affected by negation
 - Being in the syntactic scope of negation does not affect presuppositions
 - **Presupposition projection**: presuppositions are interpreted as if introduced outside the scope of the negation
 - We can use the property of projection to test for presuppositions.

“it is not the case that the chancellor decides”

↳ “*there is exactly one chancellor*, and *they do not decide*”

↳ $\exists x(\text{chancellor}(x) \wedge \forall y(\text{chancellor}(y) \leftrightarrow x = y) \wedge \neg \text{decide}(x))$

Examples of presupposition triggers

- Definite descriptions:

- *“the king of France is bald”* >> *“there is a king of France”*
- *“Mary doesn’t love her husband”* >> *“Mary has a husband”*
- *“Mary’s brother didn’t buy a house”* >> *“Mary has a brother”*

- Universal quantifiers:

- *“John kissed every girl at the party”* >> *“there were girls at the party”*

Examples of presupposition triggers

- Factive verbs (*“regret”, “realize”, “love”, “hate”, “be aware”, etc.*)
 - *“John regrets that Pola is married”* ≫ *“Pola is married”*
 - *“John realized that he was in debt”* ≫ *“John was in debt”*
- Implicative verbs (*“manage to”, “forget to”, ...*)
 - *“John forgot to close the door”* ≫ *“John intended to close the door”*
 - *“John managed to close the door”* ≫ *“John tried to close the door”*

Examples of presupposition triggers

- Aspectual verbs and items:
 - “John has stopped smoking”
 >> “John doesn’t smoke anymore”
 - “John used to fly with Lufthansa”
 >> “John doesn’t fly with Lufthansa anymore”
- It-clefts:
 - “it was John who ate the cake”
 >> “somebody ate the cake”

More presupposition projection

- Presuppositions not only “survive” negation, but also other kinds of embeddings:
 - “*the chancellor or the ministers decide*”
 >> “*there is a (exactly one) chancellor*”
 - “*John possibly regrets that Mary is married*”
 >> “*Mary is married*”
 - “*Mary believes that John has stopped smoking*”
 >> “*John used to smoke*”

Presupposition filtering

- There are contexts that can “neutralise” or **filter** some presuppositions—they block projection of these presuppositions:
 - “if John is out of town, then **his wife** is unhappy”
 >> “John has a wife” / “John is married”
 - “if John is married, then **his wife** is unhappy”
 >> ~~“John is married”~~
 - “if John is married, then **his daughter** is unhappy”
 >> “John has a daughter”

Presupposition cancellation

- In the context of negation, presuppositions can be overwritten or “cancelled” by explicitly claiming that they are false:
 - “John doesn’t **regret** that Mary is married. In fact, Mary has no husband, and John knows that.”
 - “It’s not the case that **the king of France** is bald. France is a republic.”
- Presuppositions are also cancelled when what is said, taken together with background assumptions (i.e. relevant world knowledge), is inconsistent with what is presupposed:
 - “John cried **before** finishing his thesis” >> “John finished his thesis”
 - “John died **before** finishing his thesis” >> ~~“John finished his thesis”~~

Presupposition cancellation: theoretical questions

- **The Projection Problem:** under what conditions does a sentence containing a presupposition trigger inherit this presupposition?
- **Presuppositions and Compositionality:** how can we explain the presuppositions of complex sentences in terms of the presuppositions of their parts?

Summary

- Presuppositions are triggered by a number of different words and linguistic constructions, including definite noun phrases
- Presuppositions behave differently than assertions in semantics construction: they are typically projected unchanged, rather than used in functional application
- Projected presuppositions can be filtered in the semantic composition process, and can be cancelled by contextual knowledge

Presuppositions and Anaphora

- Parallelism between **pronoun resolution** and **presuppositional filtering**
- Pronoun resolution:
 - *“John owns a donkey. He beats it.”*
 - *“If John owns a donkey, he beats it.”*
 - *“Either John does not own a donkey or he beats it.”*
- Presuppositional filtering:
 - *“Jack has children. All of Jack's children are bald.”*
 - *“If Jack has children, then all of Jack's children are bald.”*
 - *“Either Jack has no children or all of Jack's children are bald.”*

Presuppositions and Anaphora

- Presuppositions are anaphors, i.e. they want to bind to previously established discourse referents (van der Sandt, 1992)
 - *“Pedro owns a donkey. Jane knows {it / that Pedro owns a donkey}”*
 - *“if Pedro owns a donkey, Jane knows {it / that Pedro owns a donkey}”*
- Crucial difference between presuppositions and pronouns:
 - If a pronominal anaphor does not find a proper antecedent, interpretation of the sentence fails
 - If a presuppositional anaphor does not find a proper antecedent, its information is **accommodated** (added to the information state)
 - ***“they are all bald”*
 - *“all of Jack's children are bald”*

Presuppositions in DRS: basic principles

- Introduce **α -DRSs** as a new type of complex condition
- DRS construction proceeds in two steps:
 - (i) The construction rules for definite noun phrases introduce α -DRSs—this yields a “proto-DRS”
 - (ii) The α -DRSs are resolved by means of binding and accommodation—this translates a proto-DRS into a standard DRS (with a model-theoretic interpretation)

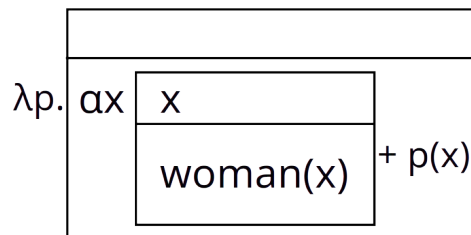
Syntax for proto-DRSs

- **Proto-DRS:** a triple (U_K, C_K, A_K) such that:
 - U_K is a set of discourse referents
 - C_K is a set of (atomic or complex) conditions
 - A_K is a set of “anaphoric” (α -)DRSs of the form $\alpha z K'$, where z is a discourse referent and K' is a proto-DRS
- A DRS is a proto-DRS (U_K, C_K, A_K) with $A_K = \emptyset$

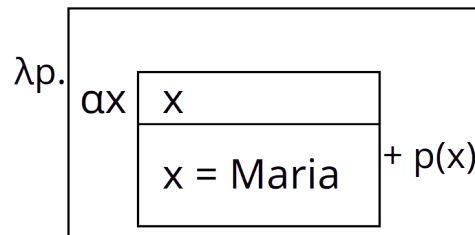
Definite Noun Phrases in DRT

- The DRS construction rules for all definite noun phrases introduce α -DRSs:

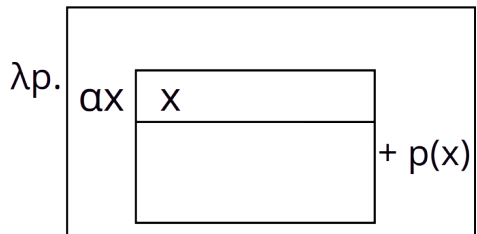
Definite descriptions (“*the woman*”):



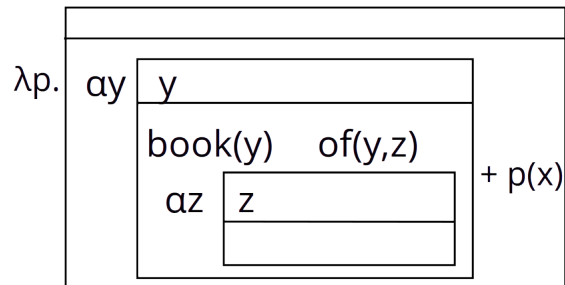
Proper names (“*Maria*”):



Pronouns (“*she*”):



Possessives (“*his book*”):

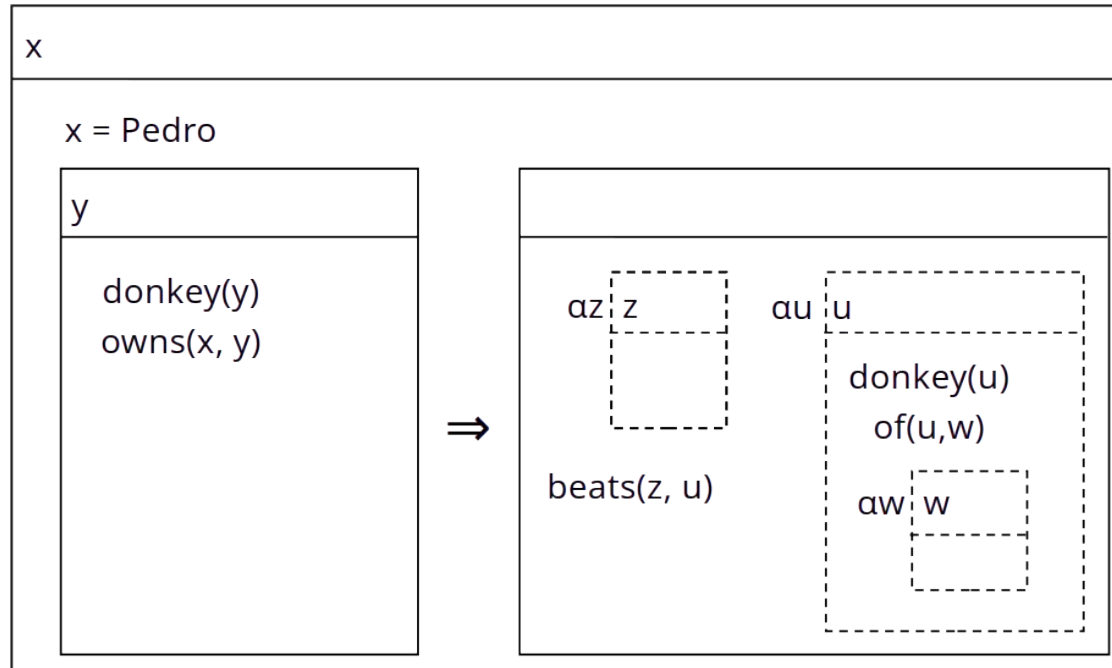


Resolution by binding

- Let K, K', K_t be some DRSs such that $K' \leq K, K_t \leq K$, and:
 - $\eta = \alpha x K_s \in A_{K'}$ (K_s is α -free)
 - $y \in U_{K_t}$ is a discourse referent that is accessible and suitable for η
- **Binding:**
 - $A_{K'} \leftarrow A_{K'} - \{\eta\}$
 - $U_{K_t} \leftarrow U_{K_t} \cup U_{K_s}$
 - $C_{K_t} \leftarrow C_{K_t} \cup C_{K_s} \cup \{x = y\}$
- Note: because K_s must be α -free, complex α -DRSs are always resolved from the inside out

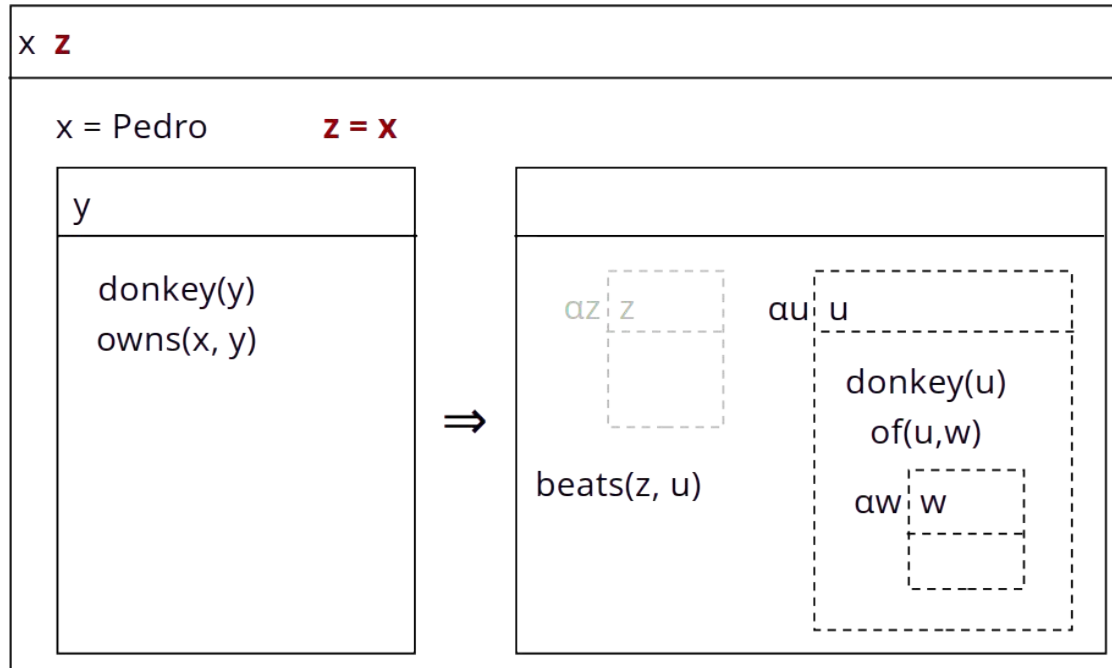
Resolution by binding: example

“if Pedro owns a donkey, he beats his donkey”



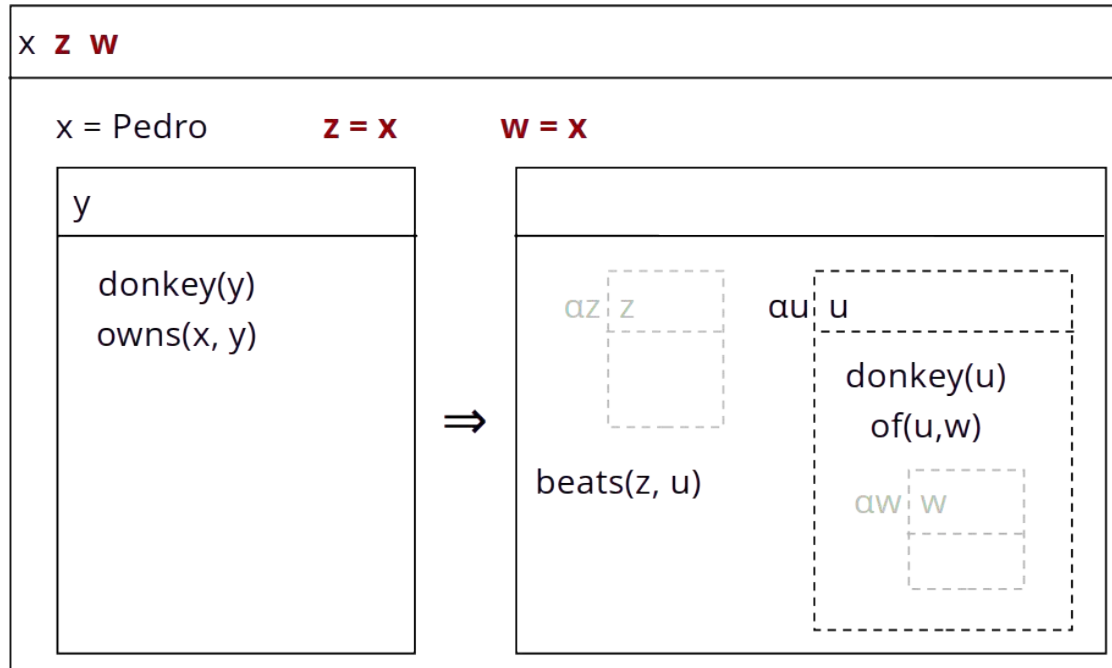
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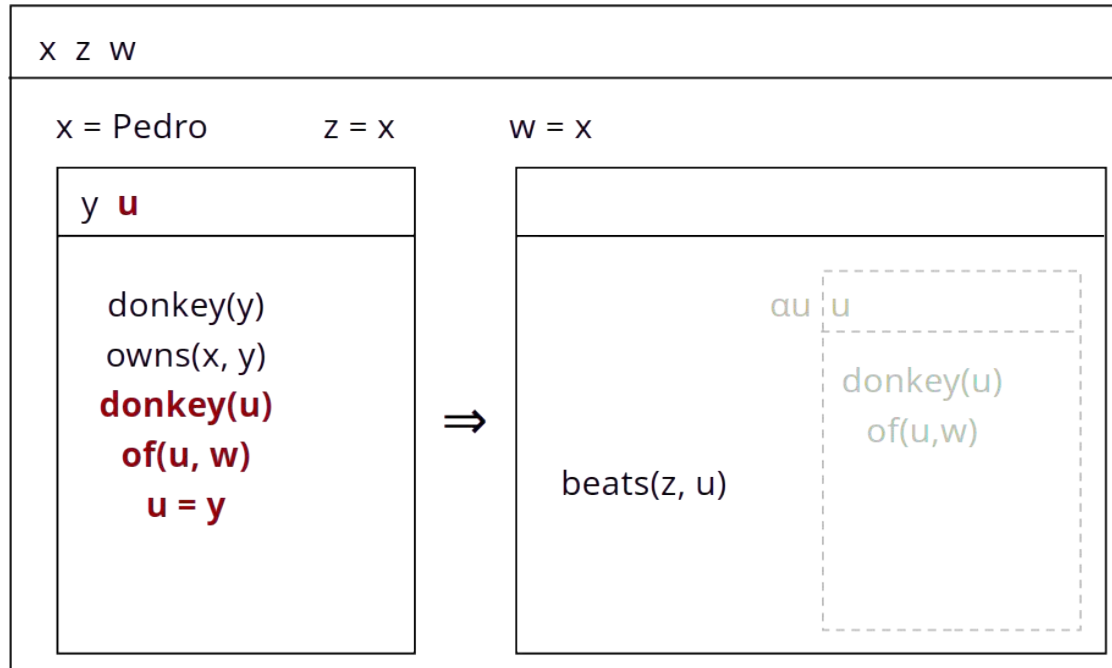
Resolution by binding: example

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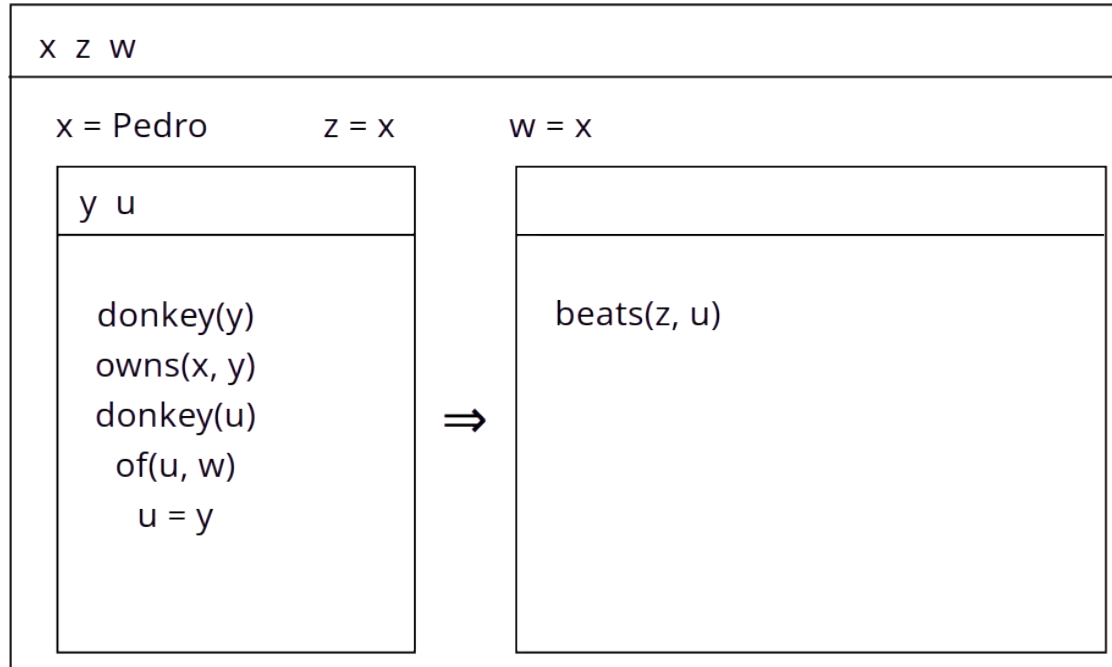
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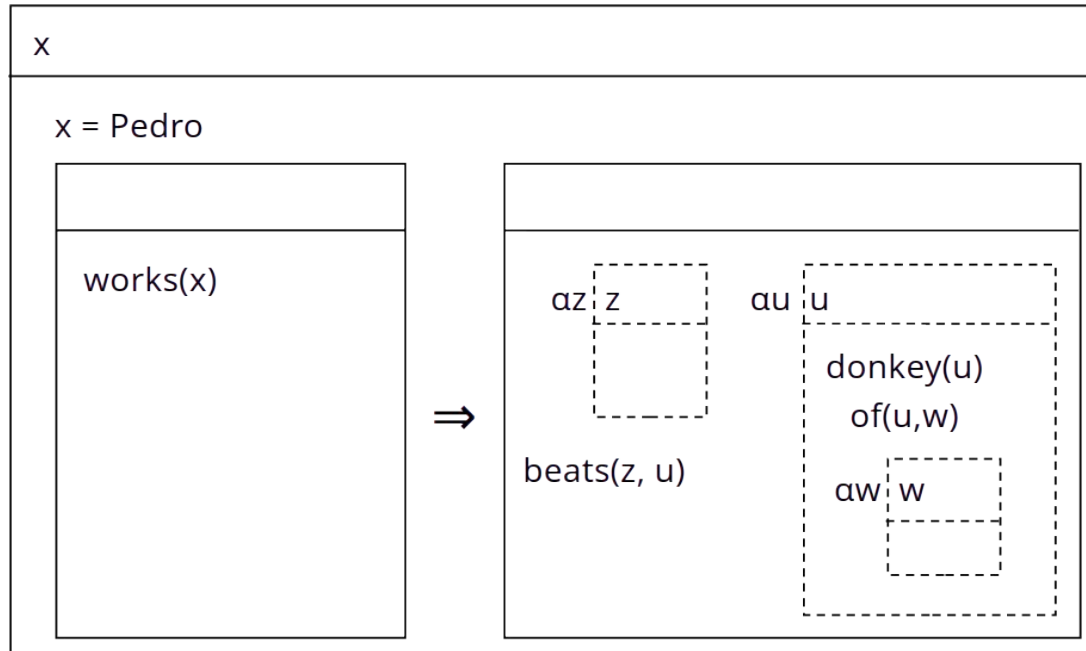


Presupposition resolution by accomodation

- Let K , K' , K_t be some DRSs such that $K' \leq K$, $K_t \leq K$, and:
 - $\eta = \alpha x K_s \in A_{K'}$ (K_s is α -free)
 - K_t is a DRS that is accessible for η
- **Accommodation:**
 - $A_{K'} \leftarrow A_{K'} - \{\eta\}$
 - $U_{Kt} \leftarrow U_{Kt} \cup U_{Ks}$
 - $C_{Kt} \leftarrow C_{Kt} \cup C_{Ks}$

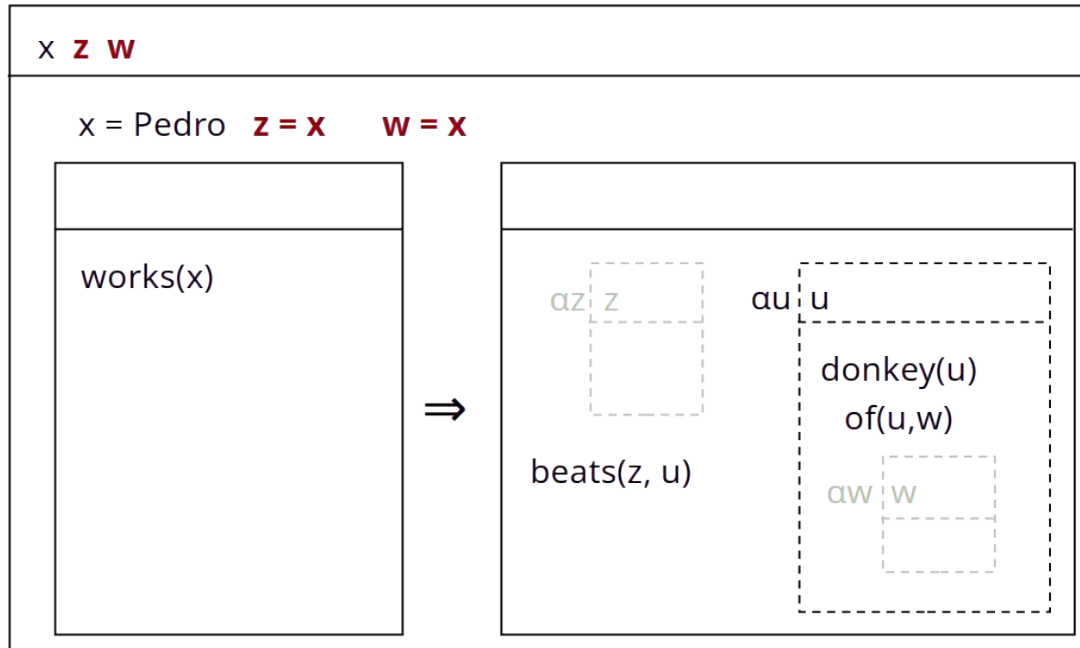
Resolution by accomodation: example

“if Pedro works, he beats his donkey”



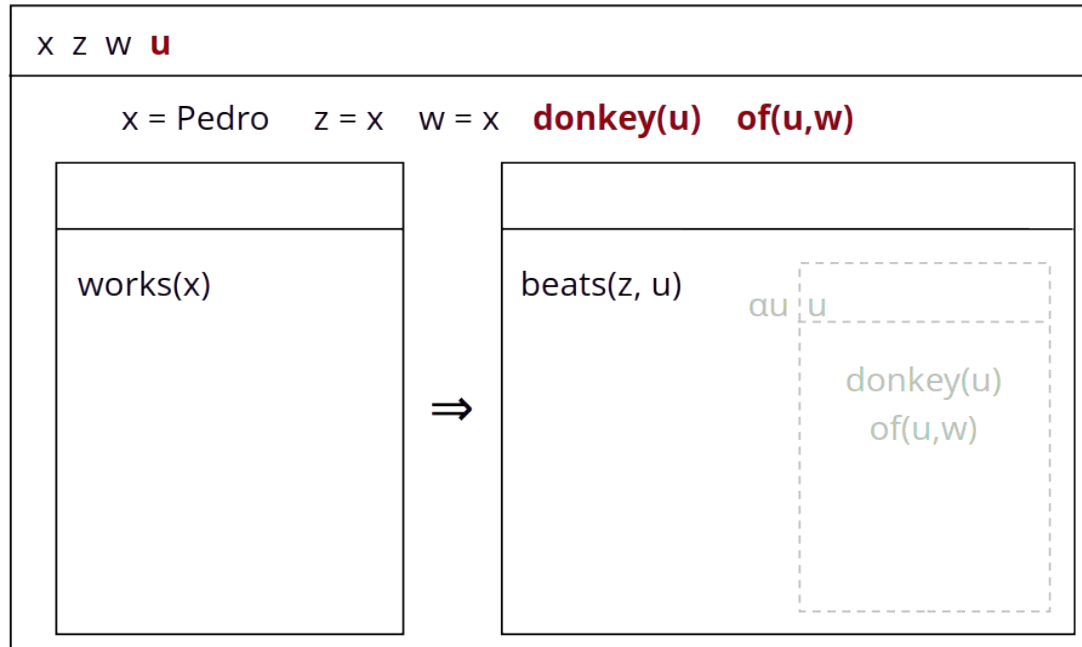
Resolution by accomodation: example

“if Pedro works, he beats his donkey”



Resolution by accomodation: example

“if Pedro works, he beats his donkey”



Presupposition resolution: preference principles

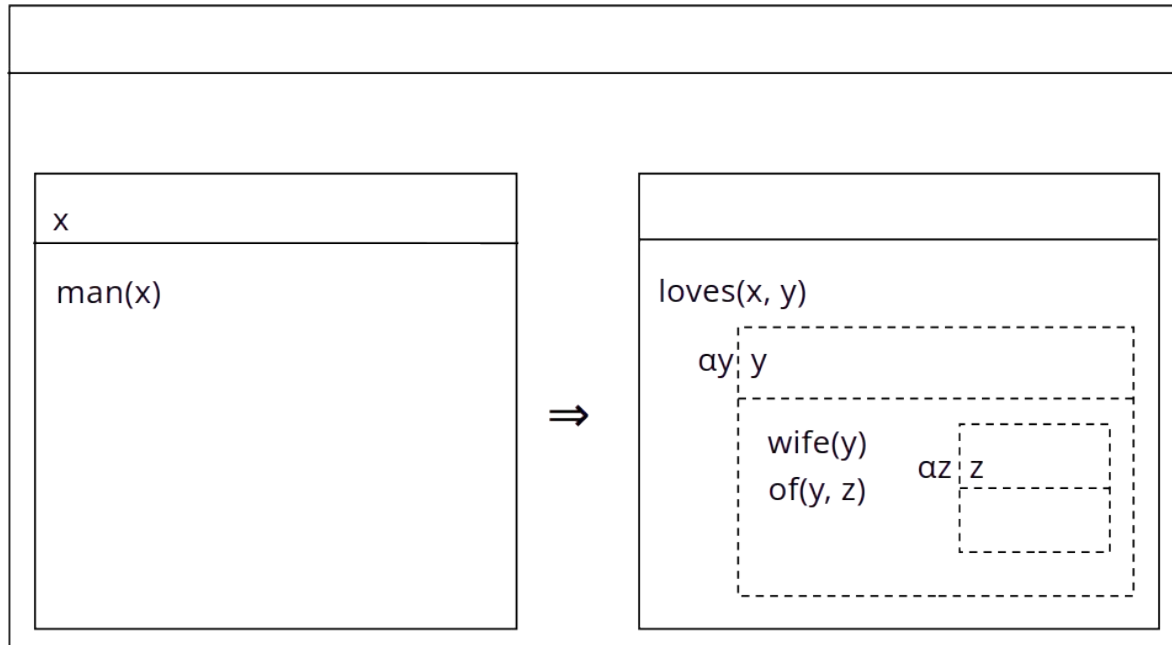
- Presupposition resolution in DRT is guided by the following principles:
 - Binding is preferred over accommodation
 - Binding works “upwards” along the accessibility relation: the closest possible antecedent is preferred
 - Accommodation works “downwards” along the accessibility relation: the highest possible DRS is preferred

Constraints on projection

- **Free variable constraint:** a resolved DRS may not contain any free discourse referents
- **Consistency and informativity constraints:** the resolved DRS must be consistent and informative

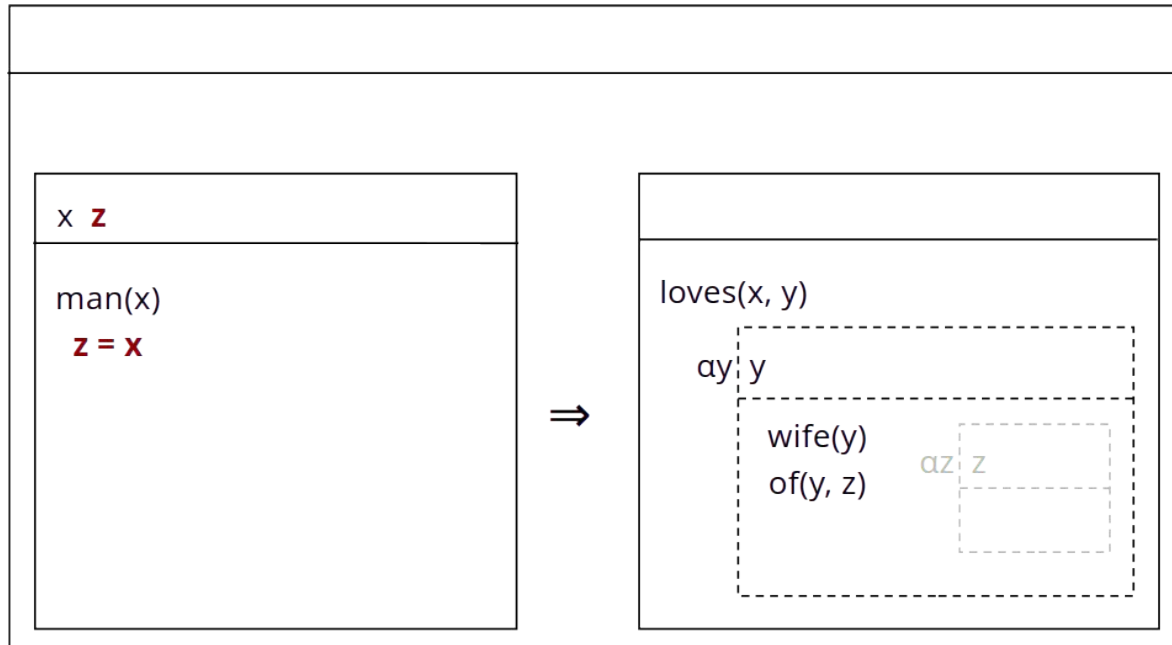
Free variable constraint: example

“every man loves his wife”



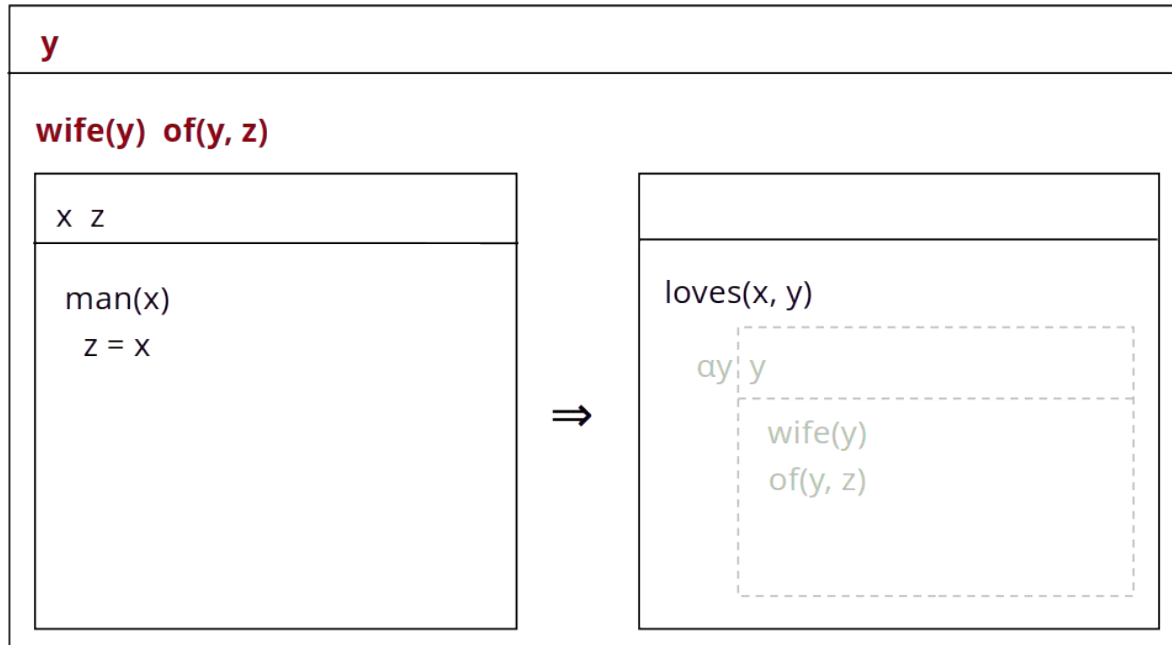
Free variable constraint: example

“every man loves his wife”



Free variable constraint: example

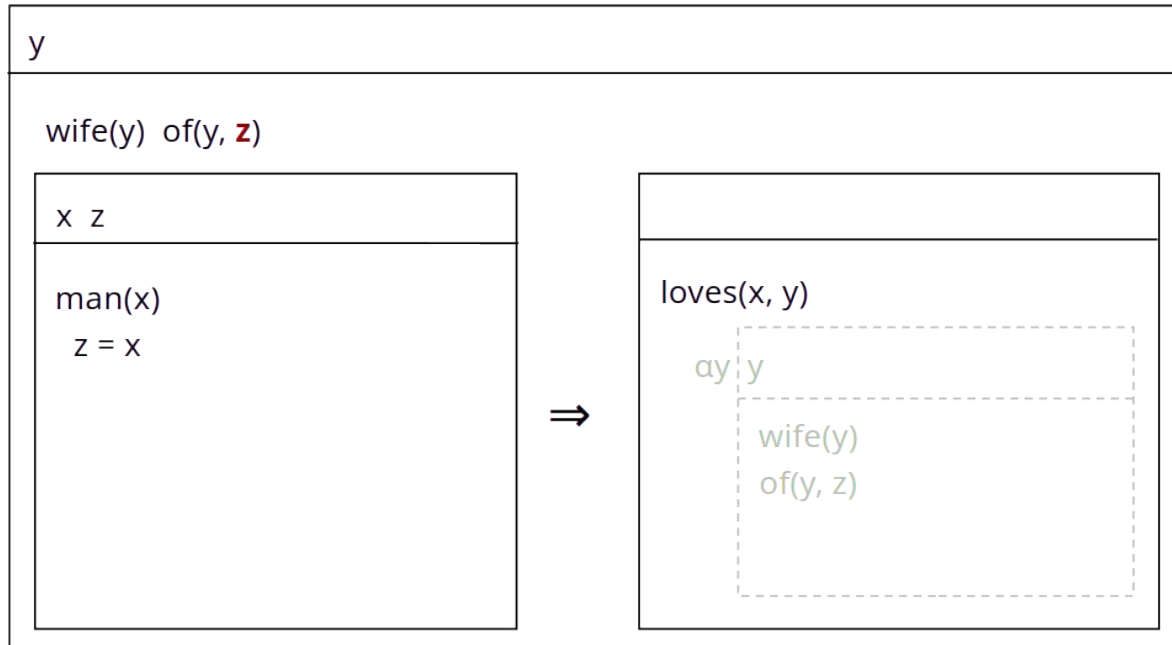
“every man loves his wife”



Free variable constraint: example

Invalid resolution:
discourse referent z is free

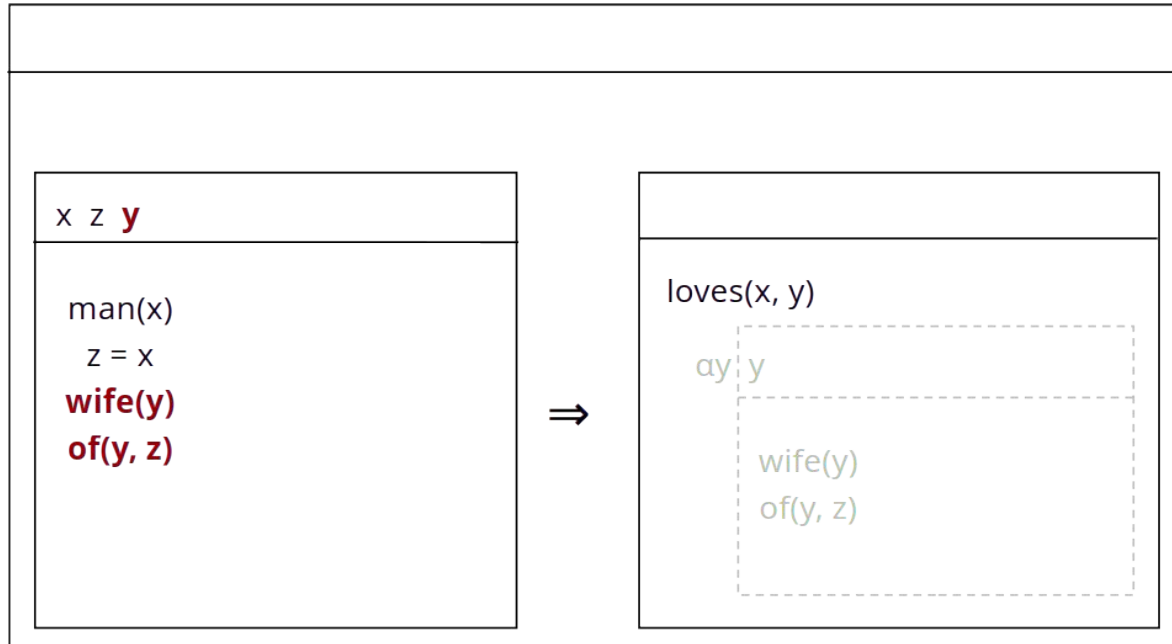
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Free variable constraint: example



“every man loves his wife”

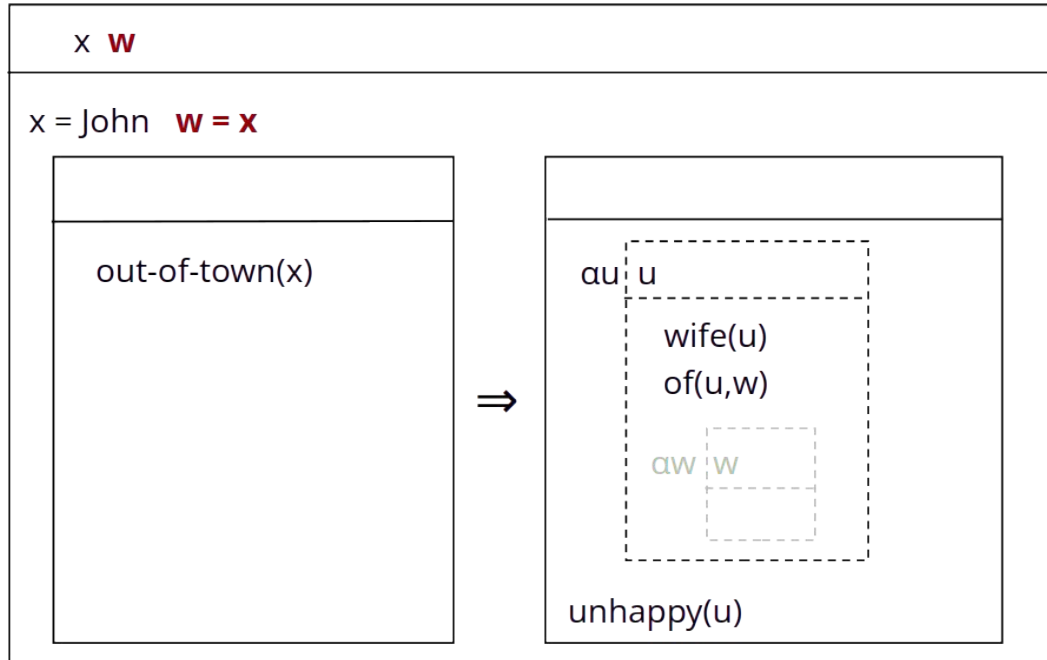


Consistency and Informativity constraints

- **Consistency:** the resolved DRS must be satisfiable (taking background knowledge into account)
 - **Local consistency:** no sub-DRS can be inconsistent with any superordinate DRS
- **Informativity:** the resolved DRS should *not* be entailed by our background knowledge
 - **Local informativity:** no sub-DRS can be entailed by any superordinate DRS

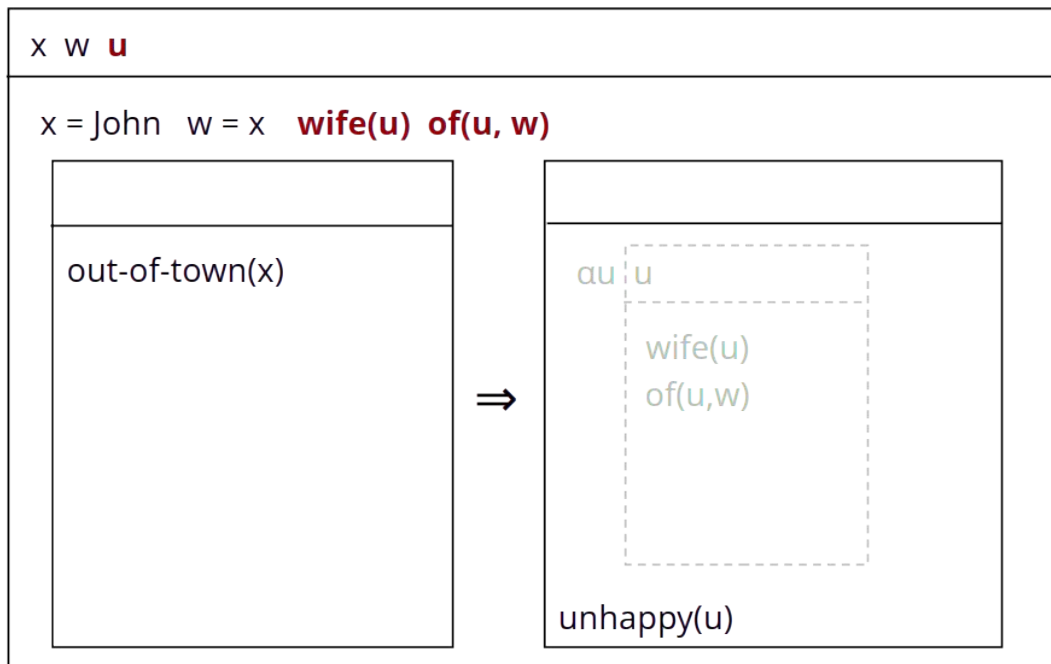
(Local) Informativity constraint: example

“if John is out of town, his wife is unhappy” $\gg$ *“John has a wife”*



(Local) Informativity constraint: example

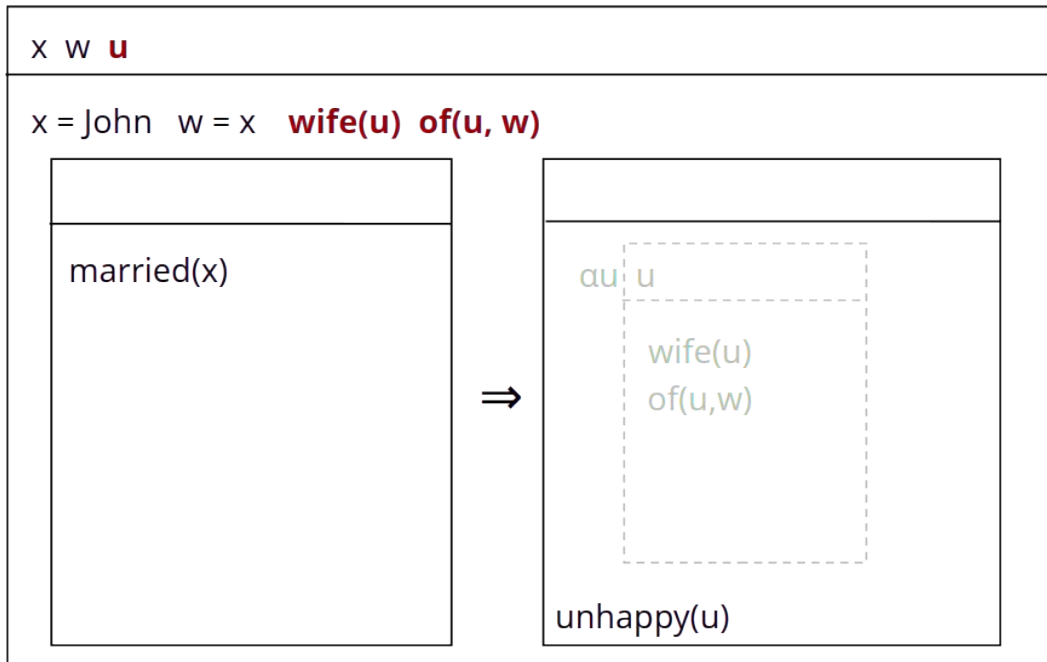
“if John is out of town, his wife is unhappy” $\gg$ *“John has a wife”*



✓ The resolved DRS
entails that John has a wife

(Local) Informativity constraint: example

“if John is married, his wife is unhappy” $\gg$ ~~“John has a wife”~~



✗ Inadmissible: the top-level DRS entails the left-hand side of the conditional ($\Rightarrow$)

(Local) Informativity constraint: example

“if John is married, his wife is unhappy” $\gg$ *~~“John has a wife”~~*

